Histology of Thyroid & Parathyroid Glands

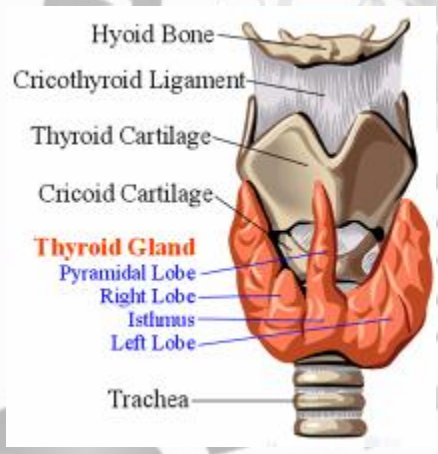
Learning Objectives

At the end of the lecture, student should be able to,

- Know the important histological features of thyroid gland.
- Discuss the capsules of thyroid gland
- Discuss the histological & functional unit of thyroid gland
- Discuss the microscopic features of cells of thyroid gland
- Discuss the histological & functional unit of parathyroid glands.
- Know the microscopic features of cells of parathyroid glands

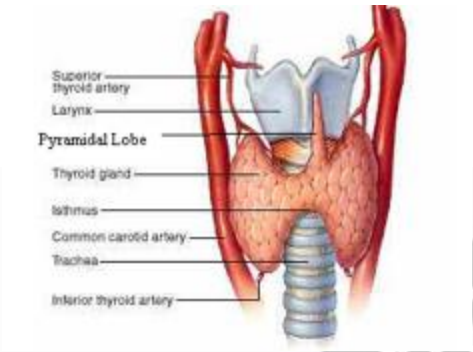
Gross Features of Thyroid Gland

- The thyroid gland is composed of two lobes connected with the isthmus.
- The organ is situated on the anterior side of the neck, lying against and around the larynx and trachea.



Pyramidal lobe of thyroid gland

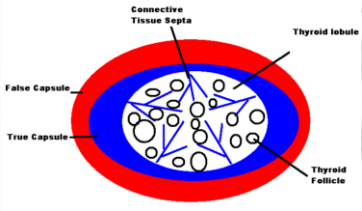
- A third lobe called the **pyramidal lobe**, frequently arises from most commonly the left lobe and ascends as far as the hyoid bone.
- It represent the lower remaining part of thyroglossal duct



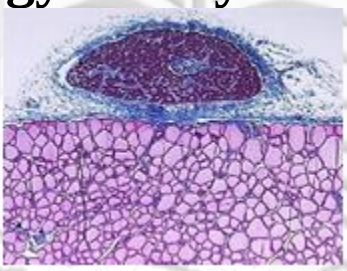
Capsules of Thyroid Gland

The thyroid gland is surrounded by the thick connective tissue sheath formed by the deep fascia of neck which is called as the **False Capsule.**
Beneath the false capsule there is an another thin connective tissue sheath, the true capsule.

- From **the true capsule** the delicate connective septa penetrate the lobes of thyroid gland and dived the lobes into lobules.
- From the septa the secondary trabeculae are arising and further divide the into smaller units and these units consists of Thyroid follicles



Histology of Thyroid Gland



The thyroid gland consists of parenchyma which is derived from endoderm and stroma which is developed from mesenchyme.

- The parafollicuar cells are ectodermal in origin and developed from neural crest cells

Microscopic Appearance

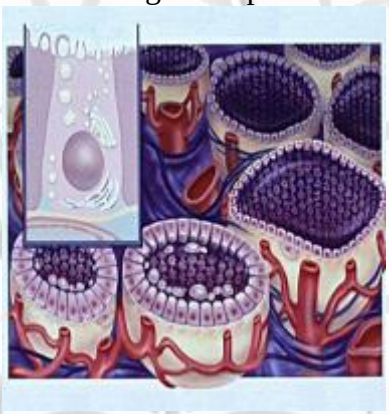
- Thyroid tissue is composed of 20–30 million microscopic spheres called **thyroid follicles.**

- The follicles are lined by a simple epithelium and their central cavity contains a gelatinous substance called **olloid.**

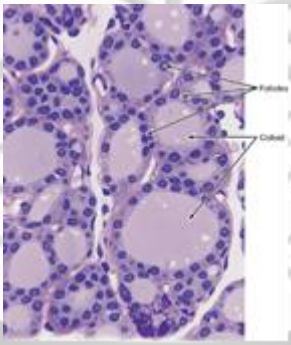


Thyroid Follicle

- The thyroid epithelium rests on a basal lamina.
- Its cells exhibit the characteristics of a cell that simultaneously synthesizes, secretes, absorbs, and digests proteins.
- The basal part of these cells is rich in rough endoplasmic reticulum.



Follicular Cells

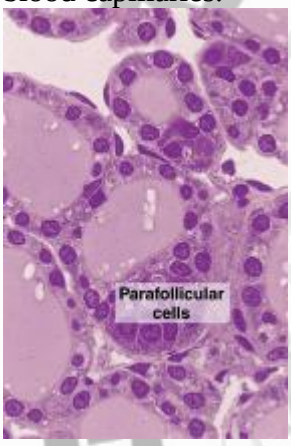


In sections, follicular cells range from **squamous to columnar** and the follicles have an extremely variable diameter.

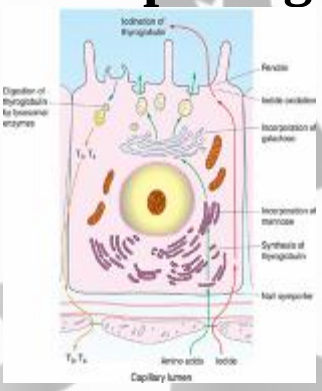
The gland is covered by a loose connective tissue capsule that sends septa into the parenchyma. As these septa gradually become thinner they reach all the follicles, separating them from one another by fine, irregular connective tissue composed mainly of reticular fibers.

- The nucleus is generally round and situated in the center of the cell.
- The apical pole has a discrete Golgi complex and small secretory granules whose content is similar to that of the follicular colloid.
- Abundant lysosomes and some large phagosomes are found in this region.
- The cell membrane of the apical pole has a moderate number of microvilli.
- Mitochondria and cisternae of rough endoplasmic reticulum are dispersed throughout the cytoplasm.

Blood Supply of Thyroid Gland

- The thyroid is an extremely vascularized organ, with an extensive blood and lymphatic capillary network surrounding the follicles.
- Endothelial cells of these capillaries are fenestrated, as they are in other endocrine glands.
- This configuration facilitates the transport of molecules between the gland cells and the blood capillaries.
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Morphological Appearance of Thyroid Follicles



- The morphological appearance of thyroid follicles varies according to the region of the gland and its functional activity.
- In the same gland, larger follicles that are full of colloid and have a cuboidal or squamous epithelium are found alongside follicles that are lined by columnar epithelium.
- Despite this variation, the gland is considered hypoactive when the average composition of these follicles is squamous.

Thyroglobulin

- In humans, there is sufficient hormone within the follicles to supply the organism for up to 3 months.
- Thyroid colloid is composed of a glycoprotein of high molecular mass (660 kDa) called **thyroglobulin.**

- Colloid fills the follicle interior, although it is often detached from the cell border because of artifacts.
- The staining of colloid is variable—sometimes basophilic and sometimes acidophilic.
- Thyroglobulin stains with periodic acid–Schiff because of its high sugar content.

Parafollicular, or C, Cell

- Another type of cell present in the thyroid, the **parafollicular,** or **C, cell,** is found as part of the follicular epithelium or as isolated clusters between thyroid follicles.
- Parafollicular cells are larger than thyroid follicular cells and with the light microscope appear less stained.

- Parafollicular cells or C cells which are located within the confines of the follicular basement lamina yet excluded from the follicle lumen by follicular epithelium cells.
- These cells **secrete calcitonin** and may be difficult to identify.
- They are somewhat larger and lighter staining than follicular cells.

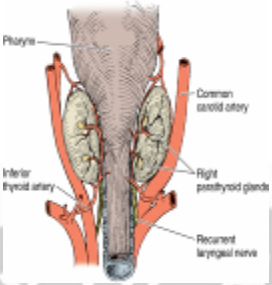
- They have a small amount of rough endoplasmic reticulum,
- long mitochondria, and
- a large Golgi complex.
- The most striking feature

of these cells is their numerous small (100–180 nm in diameter) granules containing hormone.



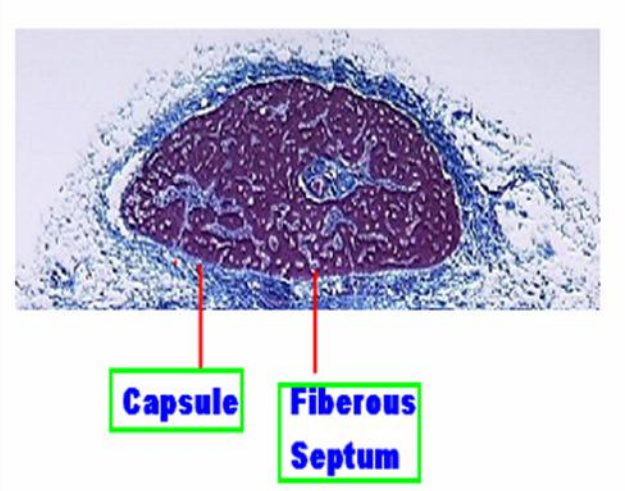
Parathyroid Glands

- The parathyroids are four small glands—3 x 6 mm—with a total weight of about 0.4 g.
- They are located behind the thyroid gland, one at each end of the upper and lower poles, usually in the capsule that covers the lobes of the thyroid.
- Sometimes they are embedded in the thyroid gland.

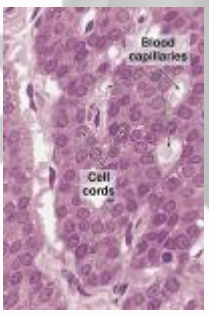


Capsule & Stroma

- Each parathyroid gland is contained within a connective tissue capsule.
- These capsules send septa into the gland, where they merge with the reticular fibers that support elongated cordlike clusters of secretory cells.



Parenchyma

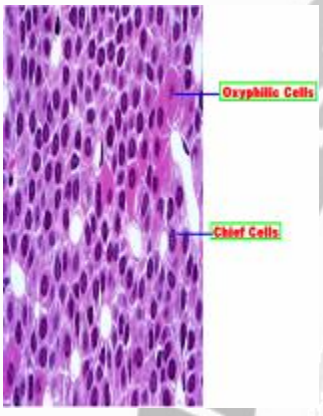


- The endocrine cells of the parathyroid are arranged in cords.
- There are two types of cells:
- **chief, or principalcells**

- **and the oxyphil cells.**

Chief or Principal Cells

- The **chief cells** are small polygonal cells
- with a vesicular nucleus and a pale-staining, slightly acidophilic cytoplasm.
- Electron microscopy shows **irregularly shaped granules** (200–400 nm in diameter) in their cytoplasm.
- They are the secretory granules containing **parathyroid hormone,** which is a polypeptide in its active form.



Oxyphil cells

- **Oxyphil cells** constitute a **smaller population.**
- **They are larger polygonal cells,** and their cytoplasm contains many acidophilic mitochondria with abundant cristae.
- The function of the oxyphil cells is not known.
- entirely lacking in small children;
- occurring first in children six to seven years old and afterwards increasing in number with age



Adipocytes.

- With increasing age, secretory cells are replaced with adipocytes.
- Adipose cells constitute more than 50% of the gland in older people.

thanks